

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

December 2012

EC402 RF & Microwave Engineering

Date: 04.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) State microwave frequency bands and its applications. [04]
- (b) Answer the following questions in short. [03]
- (1) Which element of reflex klystron replaces the output cavity of a normal klystron?
 - (2) Wave guide works as which type of filter?
 - (3) What is the primary use of TWT?
- Q - 2 (a) Based on distribution theory of transmission line, draw equivalent circuit of two wire transmission line and derive wave equation for voltage & current for a lossless transmission line. Discuss the standard assumptions. [09]
- (b) A reflex klystron is designed for a frequency of 9GHz. The anode voltage is 600V and the distance from cavity to repeller is 1 mm. Assume shunt resistance $R_{sh} = 15K\Omega$ and beam coupling is 1. The tube is oscillating at f_r at the peak of the $n=2$ mode or $1\frac{3}{4}$ mode. Find, [05]
- (1) Repeller voltage,
 - (2) Direct current necessary to give a microwave gap voltage of 200V,
 - (3) Efficiency

OR

- Q - 2 A low transmission line having characteristics impedance 50Ω is terminated by load impedance $= 100 - 150j\Omega$. Using smith chart, find [14]
- (a) Line impedance at $d=0.5\lambda$
 - (b) Line impedance at $d=0.1\lambda$
 - (c) SWR & draw SWR circle
 - (d) Distance of first minima and maxima from load
 - (e) Reflection coefficient(P)
 - (f) Return loss in db

- Q - 3 (a) Explain Faraday's rotation principle & Faraday's rotation isolator in detail [07]
- (b) Explain line impedance equations of transmission line theory. [07]

OR

- Q - 3 (a) A rectangular air-filled copper waveguide with dimension 0.9inch $\times$ 0.4inch cross section and 12" length is operated at 9.2 GHz with a dominant mode. Find [07]
- (a) Cut-off frequency,
- (b) Guide wave length,
- (c) Phase velocity,
- (d) Characteristic impedance and
- (e) The loss
- (b) Explain reflection coefficient and transmission coefficient of transmission line theory. [07]

SECTION – II

- Q - 4 (a) Why TEM mode cannot propagate in waveguide? [02]
- (b) Explain directional coupler in detail. [05]
- Q - 5 (a) Explain magic tee in detail with the help of s parameter. Also discuss about circulator. [07]
- (b) Write short note on two cavity klystron amplifier and explain electron bunching with apple gate diagram. [07]

OR

- Q - 5 (a) Derive the wave equation with complete solutions regarding wave propagation in rectangular waveguide for TM mode. [10]
- (b) Write a technical note on magnetron and explain pie mode importance. [04]
- Q - 6 (a) What is tunneling? [02]
- (b) Explain parametric amplifiers with the help of equations. [05]
- (c) Discuss the advantages and disadvantages of strip line and microstrip line. [07]

OR

- Q - 6 (a) Draw the hybrid – pi equivalent model of the common emitter configuration and discuss about equations. [07]
- (b) A microstrip line is composed of zero thickness copper conductor on a substrate having $E_r = 8.4$, $\tan \delta = 0.0005$ and thickness 2.4 mm. If the line width is 1 mm and operated at 10 GHz, calculate [07]
- (a) The characteristic impedance,
- (b) The attenuation due to conductor loss and dielectric loss
